

Howard Haonan Ho, Ph.D.

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 ORCID |  Google Scholar Profile |  Academic YouTube Channel

SUMMARY OF QUALIFICATIONS

- 15 publications in peer-reviewed journals (5), book chapter (1) and international conferences (9) including Physics of Fluids and International Symposium on Turbulence and Shear Flow Phenomena (TSFP).
- 9+ years experience in fundamental and applied research with demonstrated abilities to conduct computational thermo-fluids research involving synthetic jets, boundary layers and wakes.
- 5+ years experience in teaching undergraduate engineering courses using knowledge-centered approach to teaching and active learning strategies.
- 3+ years experience in successful research mentorship and collaboration with fellow researchers.
- 3+ years experience in professional university teaching development at the University of Toronto.

RESEARCH AND TEACHING INTERESTS

Summary of Research Interests

- Physics-Informed Neural Network
- Computational Fluid Dynamics
- Active Flow Control
- Bluff Body Aerodynamics

Summary of Teaching Interests

- Fluid Mechanics and Thermodynamics
- Aerodynamics
- Dynamics and Control Systems
- Neural Network

EDUCATION

Ph.D., Mechanical Engineering (Thermo-Fluids)

01/2022 – 06/2026

University of Toronto, Toronto, ON, Canada

Thesis: Numerical Investigation of 3D Synthetic Jet In Crossflow

Advisor: Prof. Pierre E. Sullivan & Prof. Ebenezer E. Essel

M.A.Sc., Mechanical Engineering (Thermo-Fluids)

09/2019 – 09/2021

University of Toronto, Toronto, ON, Canada

Thesis: Numerical Investigation on Flow Past A Backward Facing Step and Initial Study of Synthetic Jets

Advisor: Prof. Pierre E. Sullivan

B.Eng., Mechanical Engineering (First Class Honours)

09/2016 – 07/2019

University of Nottingham, Nottingham, United Kingdom

Thesis: An Experimental Investigation of Oil Churning and Flow Regimes Within a Model Gearbox

Advisor: Prof. Richard Jefferson-Loveday

HONOURS, AWARDS and RESEARCH FUNDING

Academic/Research Excellence Competitive Awards

Global

MDPI <i>fluids</i> Travel Award	\$900	2026
MDPI <i>fluids</i> Best Paper Award	\$500	2026

National

Digital Research Alliance of Canada (Compute Canada)

National HPC Resource Allocation (Co-applicant)	\$31,045	2026
National HPC Resource Allocation (Co-applicant)	\$40,000	2025

Institutional

University of Toronto (M.A.Sc. & Ph.D.)

Doctoral Completion Award	\$4,000	2026
Frank Howard Guest Graduate Bursary	\$4,850	2025
David Sanborn Scott Graduate Fellowship	\$9,341	2025
Thousand Island Meeting Guppie Award	Cash	2025
School of Graduate Studies Conference Grant	\$500	2023
MIE Graduate Student Conference Grants	\$900	2023

University of Nottingham (B.Eng.)

Best Individual Project on the Mechanical Engineering degree	\$200	2019
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LIST OF PUBLICATIONS AND TECHNICAL PRESENTATIONS

Books and Book Chapters

1. **Ho, H.H.**, Shirinzad, A., Essel, E.E., and Sullivan, P.E., 2025. Active Flow Control of an Airfoil in Turbulent Flow. In King, D. and Stagner, A. (editors), *Flow Turbulence Engineering In Transport*. Edited by King, D., and Stagner, A. London: Institute of Engineering Technology, ch. 2.

Refereed Journal Papers

(Published/Accepted)

5. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2026. Actuation Parameters and Boundary Layer Height Effects on a Circular Synthetic Jet in Crossflow. *Discover Fluid Mechanics*. [Accepted]
Contribution: The effects of synthetic jet momentum, actuation frequency, and boundary layer height ratio in crossflow were examined. It was summarized that a low to moderate boundary layer height ratio offered greater potential for near-wall actuation.
4. Chhetri, A., Ouedraogo, N.F., **Ho, H.H.**, Drummond, L., and Essel, E.E., 2025. Influence of Nozzle Convergence Angle on the Flow Characteristics of Synthetic Jets in a Turbulent Crossflow Boundary Layer, *ASME Journal of Fluids Engineering*. 147 (8), 081203. doi: 10.1115/1.4068035
Contribution: The effect of nozzle convergence angle on a circular synthetic jet actuator in a turbulent crossflow is studied numerically. It was found that increasing the angle enhances jet momentum, flow mixing and boundary layer penetration.
3. **Ho, H.H.**, Shirinzad, A., Essel, E.E., and Sullivan, P.E., 2024. Synthetic Jet Actuators For Active Flow Control: A Review. *MDPI Fluids*, 9 (12). doi: 10.3390/fluids9120290
Contribution: A comprehensive review explaining the working mechanism, design principles and applications of SJAs. This was an invited paper, selected as issue cover, and awarded the Best Paper award.

2. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2024. Improving 3D Synthetic Jet Modelling In Crossflow. *ASME Journal of Fluids Engineering*, 146 (3). doi: 10.1115/1.4064185

Contribution: Three different methods for modeling 3D circular SJA in a crossflow boundary layer are investigated. It was found that partial modelling of the SJA with an analytical solution can generate comparable results to a whole cavity model with dynamic mesh, reducing computational cost significantly.

1. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2022. The interactions of A Circular Synthetic Jet with A Turbulent Crossflow. *AIP Physics of Fluids*, 34 (7). doi: 10.1063/5.0099533

Contribution: The key practical impact is that engineers can choose between different jet momentum settings depending on their goals - using low momentum for wider (spanwise) coverage of separation control, or higher momentum for stronger control in the flow direction.

(Under-review/In-preparation)

1. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2026. Large Eddy Simulation of Synthetic Jet Array Flow Control on a Circular Cylinder. (*International Journal of Heat and Fluid Flow* - Invited).

Refereed Conference Papers and Abstracts

9. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2026. Large Eddy Simulation of Flow Control on a Circular Cylinder with An Array of Synthetic Jets. In 33rd Annual Conference of the Computational Fluid Dynamics Society of Canada (CSME-CFDSC-CSR 2026), Vancouver, British Columbia, Canada. https://www.youtube.com/watch?v=F_OGBEyhJ2c&t=311s
8. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2025. Improving Wind Turbine Efficiency Using Synthetic Jet Actuators. In University of Toronto Engineering Research Conference, Toronto, Ontario, Canada. <https://www.youtube.com/watch?v=f024KpFMdos>
7. **Ho, H.H.**, and Sullivan, P.E., 2025. Large Eddy Simulation of Flow Past An Infinite Span Circular Cylinder at $Re = 3900$. In Thousand Islands Fluid Mechanics Meeting, Gananoque, Ontario, Canada. <https://www.youtube.com/watch?v=jZInfJPFjxQ&t=2s>
6. **Ho, H.H.**, Machado, A., and Sullivan, P.E., 2024. Computational and Experimental Study of Spanwise Synthetic Jet Flow Control. In Thirteenth International Symposium on Turbulence and Shear Flow Phenomena (TSFP 13), Montreal, Quebec, Canada. https://www.youtube.com/watch?v=B7r_DKjtXD4&t=148s
5. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2024. Enhancing 3D Synthetic Jet Modelling for Flow Control In Aerodynamic Applications. In Thousand Islands Fluid Mechanics Meeting, Gananoque, Ontario, Canada.
4. Drummond, L., Banh, H., Ouedraogo, N.F., **Ho, H.H.**, and Essel, E.E., 2023. Effects of nozzle convergence angle on the flow characteristics of a synthetic circular jet in a crossflow. In APS Division of Fluid Dynamics, Washington DC, USA.
3. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2023. Enhanced Modelling of 3D Synthetic Jet Actuators In Crossflow. In APS Division of Fluid Dynamics, Washington DC, USA. <https://www.youtube.com/watch?v=Lzza-UVjBRc>
2. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2023. The Spanwise Effect of A Synthetic Jet In A Crossflow. In Thousand Islands Fluid Mechanics Meeting, Gananoque, Ontario, Canada.
1. **Ho, H.H.**, Essel, E.E., and Sullivan, P.E., 2022. Effects of A Varying Momentum Synthetic Jet In A Turbulent Boundary Layer. In Thousand Islands Fluid Mechanics Meeting, Gananoque, Ontario, Canada.

Symposiums

3. **Ho, H.H.**, Essel, E., and Sullivan, P.E., 2026. Large Eddy Simulation of Active Flow Control on a Circular Cylinder. In MIE Graduate Research Symposium, Toronto, Ontario, Canada.
2. **Ho, H.H.**, Essel, E., and Sullivan, P.E., 2023. Synthetic Jet Issuing Into A Crossflow. In MIE Graduate Research Symposium, Toronto, Ontario, Canada.
1. **Ho, H.H.**, and Jefferson-Loveday, R., 2019. An Experimental Investigation of Oil Churning and Flow Regimes Within a model Gearbox. In IBP Research Symposium, Nottingham, United Kingdom.

Non-Refereed Technical Presentations and Submitted Reports

1. **Ho, H.H.**, and Jefferson-Loveday, R., 2019. An Experimental Investigation of Oil Churning and Flow Regimes. Romax Technology, Nottingham, United Kingdom.

RESEARCH EXPERIENCE

Research Assistant Ph.D.

01/2022 – 06/2026

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- Performed numerical simulations using OpenFOAM to apply active flow control on a circular cylinder at transitional flow regimes, employing wall-resolved Large Eddy Simulations (LES) and Unsteady RANS (URANS).
- Development of a partial modeling of the synthetic jet actuators, resulting in comparable results to the dynamic mesh method but reducing computational cost significantly.
- Conducting collaborations both within and without my research group, specializing on working closely with experimentalist.
- Published research findings in high-profile fluid mechanics journals and conferences, including *Physics of Fluids & TSFP*

Research Assistant M.A.Sc.

09/2019 – 09/2021

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- Performed numerical simulations using OpenFOAM to analyze flow past a backward-facing step (BFS) under steady-state conditions, employing Reynolds-Averaged Navier-Stokes (RANS) and Unsteady RANS (URANS) approaches with the Launder-Sharma low-Reynolds-number $k - \epsilon$ turbulence model.
- Investigated the interaction between a synthetic jet actuator (SJA) and a turbulent boundary layer at three different momentum ratios while maintaining a constant actuation frequency to assess crossflow behavior and flow control effectiveness.
- Developed a dynamic mesh user-defined function (UDF) for diaphragm motion modeling in synthetic jet actuators, validating numerical results against experimental data to ensure accuracy and reliability.
- Submitted a thesis on the numerical investigation of synthetic jet interactions in turbulent flow in September 2021 and successfully defended the work through an oral examination.

Research Assistant B.Eng.

06/2018 – 06/2019

Department of Mechanical, Materials and Manufacturing Engineering, University of Nottingham, United Kingdom

- Conducted an experimental investigation on oil churning losses and flow regimes within a model gearbox, analyzing the effects of operating conditions on power losses and lubrication performance.

- Designed and implemented a test rig to visualize and quantify churning losses using high-speed imaging and torque measurements, providing insights into fluid dynamics and energy dissipation within the gearbox.
- Research finding is presented to industry project sponsor – Romax Technology, leading to a Ph.D opportunity
- Awarded Best Project by the department and the Institution of Mechanical Engineers (IMechE, UK) in recognition of outstanding research contributions and experimental methodology.

MENTORSHIP EXPERIENCE

Graduate Research Assistant

05/2023 – Present

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- Mentored 2 NSERC USRA students within and outside my research group to successfully conduct URANS simulations of wide scope of turbulent flows including Ahmed bodies, separated flow and synthetic jets.
- Peer-reviewed journal papers, conference paper and thesis written by the students and co-authored some of the publications

PROFESSIONAL AND TEACHING DEVELOPMENT

Perspective Professor In Training – PPIT

09/2024 – 08/2025

University of Toronto, Canada

Teaching Engineering in Higher Education – TEP 1203H

09/2024 – 12/2024

University of Toronto, Canada

Research Method and Project Execution – TEP 5500H

09/2023 – 12/2023

University of Toronto, Canada

TEACHING EXPERIENCE

Course Instructor (Full-Term)

Dynamics MIE 100H – University of Toronto

01/2024 – 05/2024

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- **Details:** First Year Undergraduate dynamics Course
- **Logistics:** 650 Undergraduate Students (Class of 120)
- **Evaluation of Instructor:** score: 3.9/5.0 (Department Mean 3.7/5.0)

Guest Lecturer

Conservation of Momentum – University of Toronto

10/2025

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- 1-hour lecture on the conservation of momentum in a 3rd year fluid course (MIE 312).

Introduction On Synthetic Jet Actuators – Concordia University

06/2023

Department of Mechanical Engineering, Concordia University, Canada

- 1-hour introductory lecture on the synthetic jet actuator, an active flow control device with great potential in suppressing flow separation and improving aircraft fuel economy.

Workshop Host

Introduction to Paraview 5.10 – Concordia University

06/2023

Department of Mechanical Engineering, Concordia University, Canada

- 2-hours hands-on workshop using the open-source post-processing software to analyze fluid simulation result.

Teaching Assistant

HVAC Fundamentals MIE 507H – University of Toronto

09/2025 – 12/2026

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- Level 500 course (4th year Undergraduate/ postgraduate), average class of 20 students.

Fluid Mechanics MIE 312H – University of Toronto

09/2024 – 12/2026

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- Third Year Undergraduate Course, average class of 60 students.

Dynamics MIE 100H – University of Toronto

01/2021 – 05/2026

Department of Mechanical and Industrial Engineering, University of Toronto, Canada

- First Year Undergraduate Course, average class of 70 students.

PROFESSIONAL AND COMMUNITY SERVICES

Reviewer

International Fluid Mechanics Journals

2023 – Present

- Physics of Fluids
- Journal of Flow Visualization

Academic Evaluation

Teaching Evaluation for Faculty Promotion Case

2025

- UTIAS, University of Toronto

Membership

- Member, American Society for Mechanical Engineering (ASME) **2023 – Present**
- Member, AIAA **2023 – Present**